

Recoup (CA 1)

SYNOPSIS

**Demand Forecasting in Retail using Multiple
Regression and GenAI**

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Course Code: CSM355

Date of Submission: 28-04-2026



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Introduction:

Background/Context:

The retail industry is one of the most data-intensive sectors, generating massive volumes of transactional, customer, and inventory data on a daily basis. Accurate demand forecasting is crucial for retailers to maintain optimal stock levels, reduce wastage, improve customer satisfaction, and maximize revenue. Inaccurate forecasts lead to either overstock (increasing holding costs) or stockouts (causing lost sales).

Traditional forecasting methods such as moving averages and exponential smoothing are limited in capturing complex, non-linear relationships between multiple influencing factors. With the advancement of Machine Learning (ML) and Generative AI (GenAI), it is now possible to build smarter forecasting models that incorporate historical sales data, promotions, seasonality, price, and external factors.

This project focuses on building a retail demand forecasting system using Multiple Regression and GenAI to accurately predict future product demand across different store segments.

Problem Statement:

Existing demand forecasting approaches in retail face several challenges:

- Inability to account for multiple influencing variables simultaneously.
- Poor adaptability to seasonal fluctuations, promotions, and price changes.
- Lack of interpretability in black-box ML models.
- Difficulty in generating human-readable demand insights automatically.

This project addresses these challenges by combining Multiple Regression for structured prediction with GenAI for natural language-based demand insights.

Rationale/Justification:

Retail demand forecasting has a direct impact on supply chain efficiency, financial planning, and customer experience. With increasing competition and shrinking margins, retailers need intelligent forecasting tools that go beyond simple statistical models.

This project integrates Multiple Regression (a wellunderstood statistical model) with Generative AI capabilities to both predict demand numerically and explain those predictions in

natural language. This dual approach enhances business utility and aligns with modern AI-driven retail analytics trends.

Objectives:

General Objective:

To design and implement a demand forecasting system for retail using Multiple Regression and Generative AI that predicts future product demand and generates human-readable insights.

Specific Objectives:

- To collect and preprocess a retail sales dataset.
- To perform Exploratory Data Analysis (EDA) to understand sales trends, seasonality, and variable relationships.
- To identify and handle data issues such as missing values, outliers, and multicollinearity.
- To build and train a Multiple Regression model for demand forecasting.
- To integrate GenAI (LLM) to generate natural language summaries and recommendations from forecast results.
- To evaluate model performance using metrics like RMSE, MAE, and R-squared.

- To analyze feature importance and understand key demand drivers.
- To compare forecasting accuracy with baseline methods.

Scope of the Project:

Included:

- Analysis of retail sales data across multiple product categories.
- Data preprocessing including handling missing values and feature engineering.
- Exploratory Data Analysis using visualization techniques.
- Implementation of Multiple Regression for demand prediction.
- Integration of GenAI (LLM API) for generating demand insights.
- Model performance evaluation and comparison.

Excluded:

- Real-time deployment in live retail environments.
- Integration with enterprise ERP or POS systems.
- Deep learning models such as LSTM (optional future scope).
- Handling real-time streaming sales data.

Literature Review (brief):

Demand forecasting in retail has been extensively studied using statistical, machine learning, and deep learning approaches. Classical models such as ARIMA and exponential smoothing are widely used but fail to incorporate multiple external variables simultaneously.

Multiple Regression models have proven effective in capturing the influence of promotions, price, seasonality, and store location on demand. Studies show that regression-based models outperform simple univariate forecasters when multiple predictors are available.

Recent advances in Generative AI, particularly Large Language Models (LLMs) such as GPT-4, have opened new possibilities for translating complex model outputs into actionable business insights. Research highlights that combining predictive ML

models with GenAI enhances decision-making in supply chain and retail contexts.

Benchmark datasets such as the Rossmann Store Sales dataset and M5 Forecasting dataset are commonly used in retail demand forecasting research, providing rich features including promotions, holidays, and store metadata.

Methodology:

Research Design:

This project follows a quantitative and experimental research approach. It involves analyzing labeled retail sales data and applying Multiple Regression and GenAI techniques for forecasting and insight generation.

Data Collection Methods:

- Dataset: Rossmann Store Sales Dataset. Shape of dataset is (1017209, 18)
- Features include: store ID, date, sales, customers, promotions, school/state holidays, competition distance, and store type.

Data Preprocessing:

- Removal of irrelevant features (e.g., ID columns).
- Handling missing values and zero-sales days.

- Encoding categorical variables such as store type and assortment.
- Feature engineering: extracting month, week, dayof-week from date.
- Normalization and scaling of numerical features.

Exploratory Data Analysis (EDA):

- Analysis of sales distribution across stores, days, and seasons.
- Visualization using line plots, boxplots, heatmaps, and bar charts.
- Identification of skewed features, outliers, and multicollinearity.
- Correlation analysis between sales and predictor variables.

Model Development:

- Splitting data into training and testing sets (80/20 split).
- Training a Multiple Regression model using Ordinary Least Squares (OLS).
- Generating predictions and computing residuals.

- Integrating GenAI (LLM API) to generate natural language demand summaries and recommendations from regression outputs.

Evaluation Metrics:

- Root Mean Square Error (RMSE)
- Mean Absolute Error (MAE)
- R-Squared (R^2) Score
- Mean Absolute Percentage Error (MAPE)

Tools and Technologies:

- Programming Language: Python
- Libraries: Pandas, NumPy (data handling); Scikitlearn (regression model); Statsmodels (OLS); Matplotlib, Seaborn (visualization)
- GenAI Integration: Anthropic Claude API
- Environment: Jupyter Notebook / Google Colab

Expected Outcomes/Deliverables:

- A trained Multiple Regression model capable of forecasting retail product demand.
- GenAI-generated natural language summaries of demand trends and recommendations.

- Visualization of sales trends and forecast vs actual comparison.

User Interface Design:

The project includes an interactive web-based dashboard built using Streamlit, a Python framework widely used for deploying machine learning applications. The dashboard is designed to make the demand forecasting system accessible to non-technical retail stakeholders, enabling them to interact with model outputs without any programming knowledge.

- The interface is organized into four functional modules:

1. Overview Dashboard

The landing page presents key performance indicators including average daily sales, promotion lift percentage, model R^2 score, and Mean Absolute Percentage Error (MAPE). It includes a monthly sales trend line chart illustrating seasonal demand patterns and a bar chart comparing average sales across different store types. This gives the user an

immediate snapshot of retail performance and forecast accuracy.

2. What-If Demand Simulator

This is one of the unique contributions of the project. The simulator allows users to interactively adjust input parameters such as customer footfall, promotion status, store type, competition distance, and month — and instantly observe the corresponding change in predicted demand. This feature replicates a real-world retail planning scenario where managers test different business conditions before making inventory or promotional decisions. The prediction is generated in real time using the trained Multiple Regression model running in the backend.

3. AI Analyst (GenAI Integration)

This module embeds a conversational interface powered by the Anthropic Claude API. Users can type natural language questions about the forecast data — for example, "Why is demand spiking in December?" or "Which store type performs best during promotions?" — and receive contextual, human-

readable answers generated by the large language model. The AI analyst is provided with the regression model's outputs, feature coefficients, and summary statistics as context, enabling it to generate responses that are grounded in the actual model data rather than generic responses. This represents the core GenAI integration of the project.

4. Model Evaluation Panel

This section presents the complete model evaluation report including RMSE, MAE, R^2 , and MAPE scores for both training and test sets. It also displays the feature importance chart showing standardized regression coefficients, and an Actual vs Predicted scatter plot for visual validation of model performance.

Project Timeline:

Phase	Activity
Week 1	Data Collection + Dataset Understanding + Initial Preprocessing
Week 2	EDA (Sales Trends, Heatmaps, Seasonality Analysis) + Feature Engineering
Week 3	Data Cleaning + Encoding + Handling Outliers + Multicollinearity Check
Week 4	Multiple Regression Model Training + OLS Analysis
Week 5	GenAI Integration + Generating Demand Insights + Model Evaluation
Week 6	Documentation + Final Report + Submission

Future Scope:

- Integration with real-time POS (Point-of-Sale) systems for live demand forecasting.
- Implementation of deep learning models such as LSTM for time-series forecasting.
- Deployment as a web dashboard using Flask or Streamlit.
- Expanding GenAI integration for automated inventory replenishment recommendations.
- Incorporating external factors such as weather, social media trends, and competitor pricing.

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Declaration:

I hereby declare that the project titled “Demand Forecasting in Retail using Multiple Regression and GenAI” is an original work carried out as part of the Bachelor of Technology program.

This work has not been submitted elsewhere for any degree or diploma. All sources of information have been duly acknowledged.

Signatures:

Supervisor:

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(Signature) _____